7. A school is planning to run several training days next year for its new teachers, middle leaders and senior leaders.

Next year, the school will need to run

- at least 20 training days in total,
- at most twice as many training days for new teachers when compared to the total number of training days required for both middle and senior leaders,
- at most 25% of the training days for senior leaders.

The costs of running a training day for new teachers, middle leaders and senior leaders are £400, £550 and £750 respectively.

The school wants to minimise the total cost of running the training days.

Let x be the number of training days required for new teachers.
Let y be the number of training days required for middle leaders.
Let z be the number of training days required for senior leaders.

(a) Formulate this information as a linear programming problem. State the objective and list the constraints as simplified inequalities with integer coefficients.

(6)

The school decides that the number of training days for middle leaders and the number of training days for senior leaders should be in the ratio 5 : 3
This reduces two of the constraints to $5x \leqslant 16y$ and $4y \leqslant 5x$.

(b) (i) Express the constraint representing the requirement for a total of at least 20 training days as a simplified inequality with integer coefficients in terms of x and y only.

(ii) Express the objective in the form $ax + by$ where a and b are integers.

(3)

(c) Represent the constraints in x and y only on Diagram 1 in the answer book. Hence determine, and label, the feasible region, R.

(4)

(d) Use the objective line method to locate the optimal vertex, V, of the feasible region. You must make your objective line clear and label the optimal vertex, V.

(3)

(e) Hence, determine

(i) the total cost of running the training days,

(ii) the number of training days required for senior leaders.

(2)

(Total 18 marks)

TOTAL FOR PAPER: 75 MARKS

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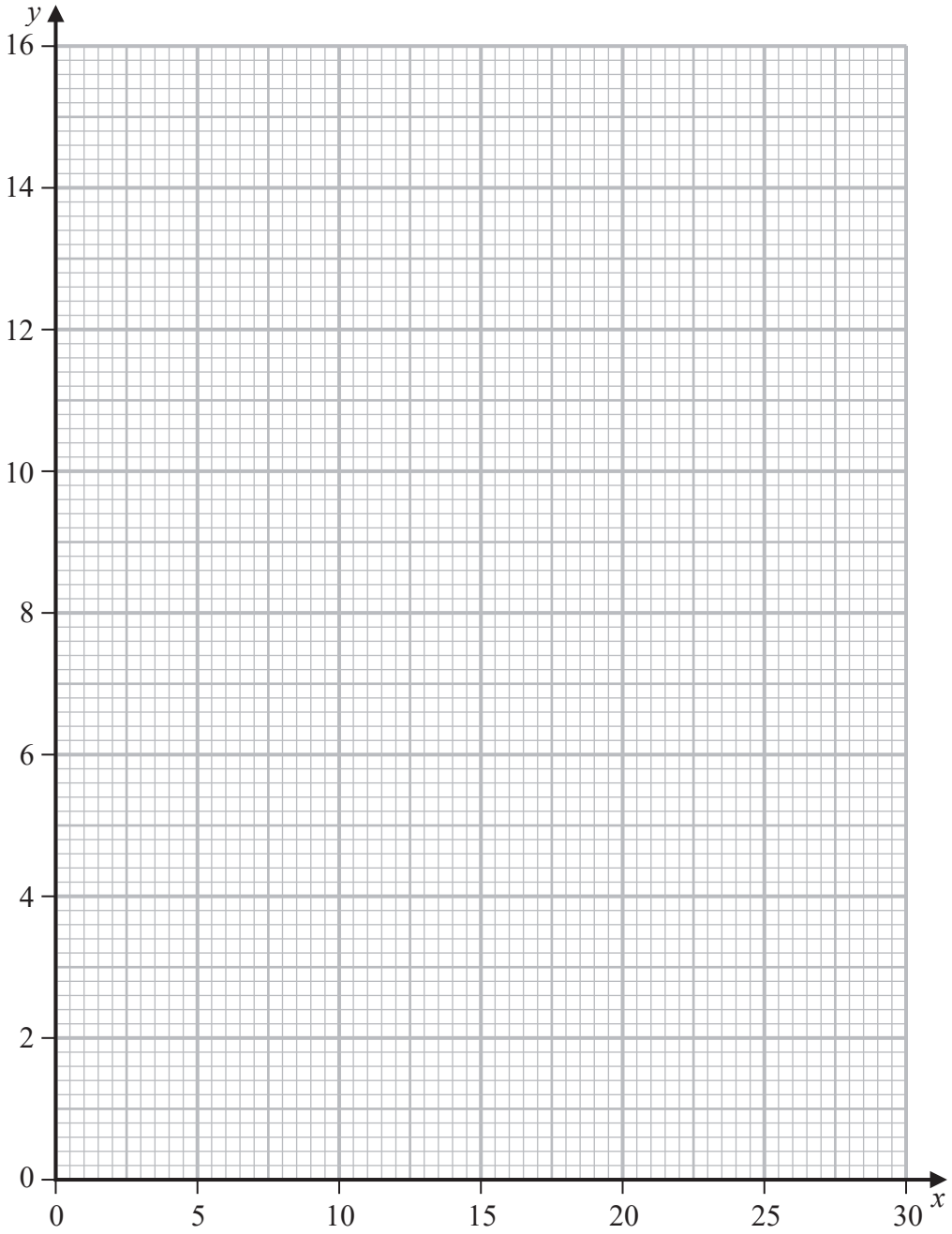


Diagram 1

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Question 7 continued

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(Total 18 marks)

TOTAL FOR PAPER: 75 MARKS

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2. A restaurant sells two sizes of pizza, small and large. The restaurant owner knows that, each evening, she needs to make

- at least 85 pizzas in total
- at least twice as many large pizzas as small pizzas

In addition, at most 80% of the pizzas must be large.

Each small pizza costs £2 to make and each large pizza costs £3 to make.

The restaurant owner wants to minimise her costs.

Let x represent the number of small pizzas made each evening and let y represent the number of large pizzas made each evening.

Formulate the information above as a linear programming problem. State the objective and list the constraints as simplified inequalities with integer coefficients. You should **not** attempt to solve the problem.

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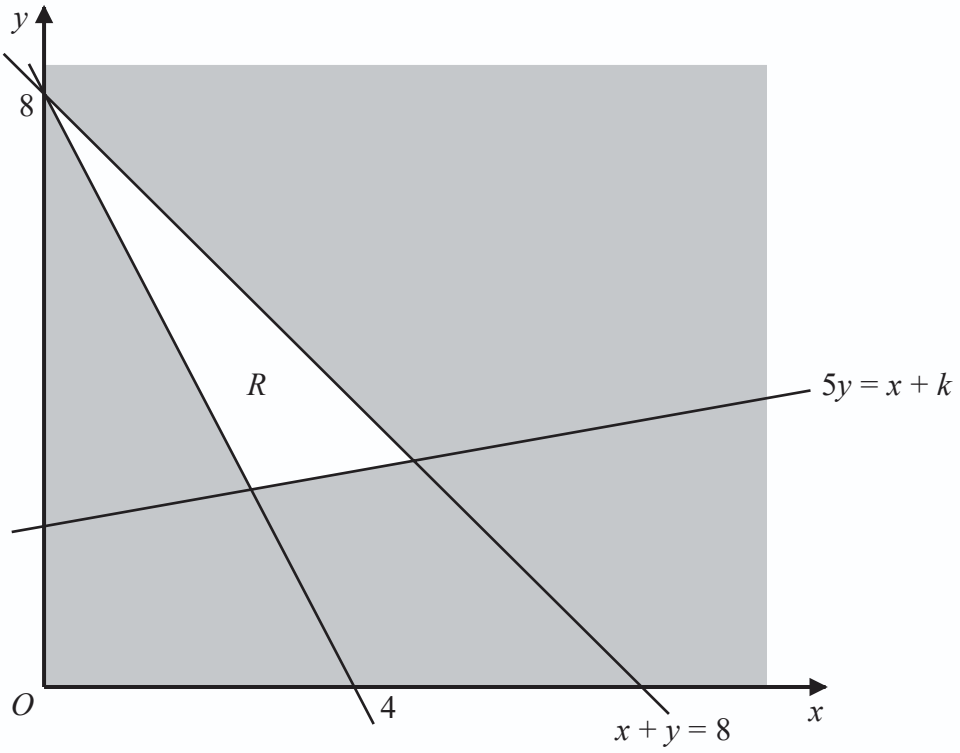


Figure 3

Figure 3 shows the constraints of a linear programming problem in x and y, where R is the feasible region. The equations of two of the lines have been shown in Figure 3.

Given that k is a positive constant,

(a) determine, in terms of k where necessary, the inequalities that define R.

(4)

The objective is to maximise $P = 5x + ky$

Given that the value of P is 38 at the optimal vertex of R,

(b) determine the possible value(s) of k. You must show algebraic working and make your method clear.

(7)

(Total 11 marks)

TOTAL FOR PAPER: 75 MARKS

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Question 7 continued

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7. A company makes three types of storage container, small, medium and large.

The company owner knows that each week she should make

- at least 40 containers in total
- at least twice as many large containers as medium containers
- at most 60% small containers

Each small container requires 1 hour to make, each medium container requires 1.5 hours to make, and each large container requires 2.5 hours to make. The company has a total of 75 hours per week available to make all the containers.

Each small container costs £9 to make, each medium container costs £12 to make and each large container costs £16 to make.

The company owner wants to minimise her total cost.

- Let x represent the number of small containers made
- Let y represent the number of medium containers made
- Let z represent the number of large containers made

(a) Formulate this information as a linear programming problem. State the objective and list the constraints as simplified inequalities with integer coefficients.

(7)

The company owner now decides to make exactly 45 containers.

(b) Explain why the minimum total cost is achieved when $7x + 4y$ is maximised.

(3)

The requirement to make exactly 45 containers reduces the constraints of the problem to the following:

$$x + 3y \leqslant 45$$

$$0 \leqslant x \leqslant 27$$

$$3x + 2y \geqslant 75$$

$$y \geqslant 0$$

(c) Represent these constraints on Diagram 1 in the answer book. Hence determine, and label, the feasible region, R.

(4)

(d) Use the objective line method to find the optimal vertex, V, of the feasible region. You must make your objective line clear and label V.

(2)

(e) Write down the number of each type of container the company should make. Calculate the corresponding total cost.

(2)

(Total 18 marks)

TOTAL FOR PAPER = 75 MARKS

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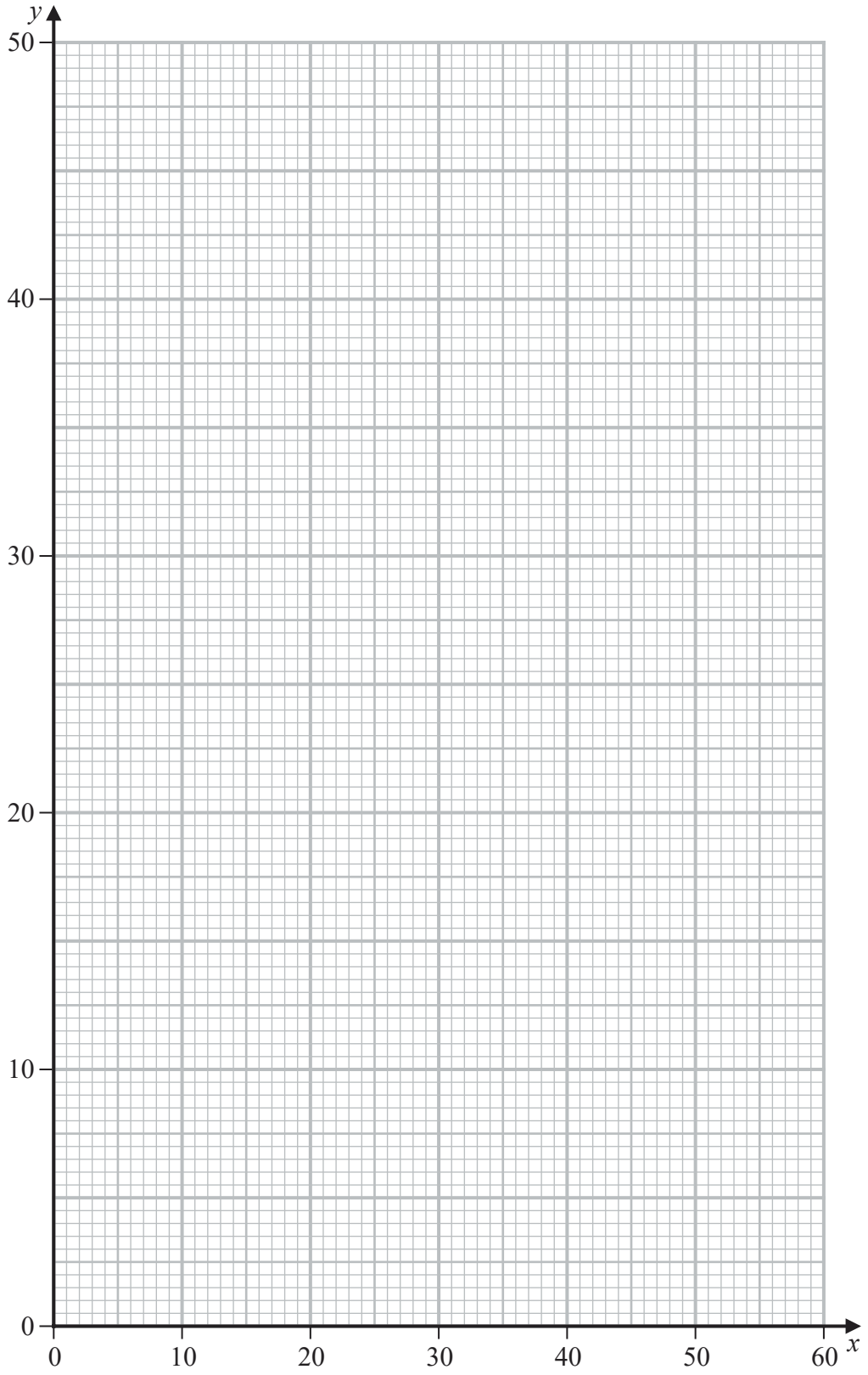


Diagram 1

Question 7 continued

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6. Martin is making three types of cake for a picnic. The three types of cake are carrot cake, apple cake and chocolate cake. Along with other ingredients,

- each carrot cake contains 275 grams of flour, 300 grams of sugar and 5 eggs
- each apple cake contains 200 grams of flour, 400 grams of sugar and 2 eggs
- each chocolate cake contains 100 grams of flour, 400 grams of sugar and 3 eggs

If Martin makes only one type of cake then he has enough time to prepare 15 carrot cakes or 20 apple cakes or 30 chocolate cakes.

Martin has 5.5 kilograms of flour and 70 eggs available and he has promised the picnic organisers that he will make at least 18 cakes in total.

Martin plans to make a selection of these cakes and wants to minimise the total amount of sugar that he uses.

Let x be the number of carrot cakes made, y the number of apple cakes made and z the number of chocolate cakes made.

(a) Formulate this information as a linear programming problem. State the objective and list the constraints as simplified inequalities with integer coefficients.

(6)

A further constraint is that $y = 2z$

(b) Explain what this constraint means in the context of the question.

(1)

The constraint $y = 2z$ reduces the problem to the following

Minimise $P = 300x + 600y$

subject to $11x + 10y \leqslant 220$

$10x + 7y \leqslant 140$

$x + y \leqslant 15$

$2x + 3y \geqslant 36$

$x \geqslant 0,\ y \geqslant 0$

(c) Represent these constraints on Diagram 1 in the answer book. Hence determine, and label, the feasible region, R.

(4)

(d) Use the objective line method to find the optimal number of each type of cake that Martin should make, and the amount of sugar used.

(4)

(e) Determine how much flour and how many eggs Martin will have left over after making the optimal number of cakes.

(2)

(Total for Question 6 is 17 marks)

TOTAL FOR PAPER: 75 MARKS

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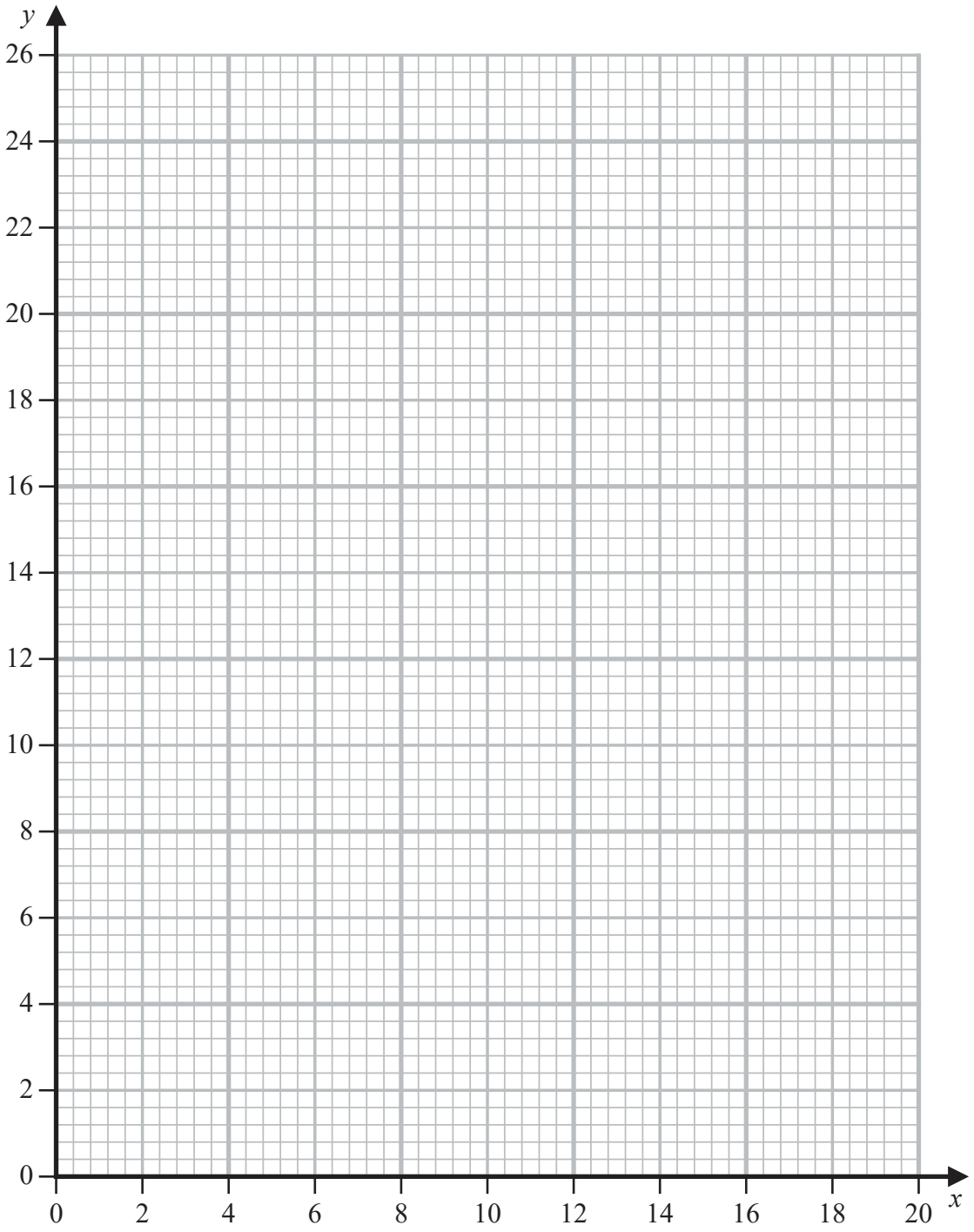


Diagram 1

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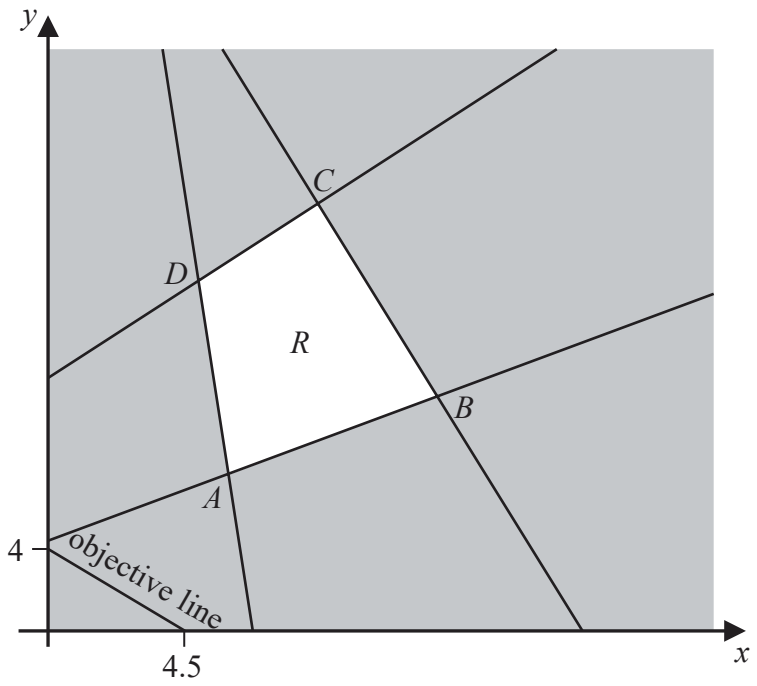


Figure 3

Figure 3 shows the constraints of a linear programming problem in x and y.
The unshaded area, including its boundaries, forms the feasible region, R.

The four vertices of R are $A(6, 8)$, $B(13, 12)$, $C(9, 22)$ and $D(5, 18)$.

An objective line has been drawn and labelled on the graph.

When the objective function, P, is maximised, the value of P is 540

When the objective function, P, is minimised, the value of P is k

Determine the value of k. You must make your method and working clear.

(You may assume that the objective function, P, takes the form $ax + by$ where a and b are constants.)

(Total for Question 5 is 5 marks)

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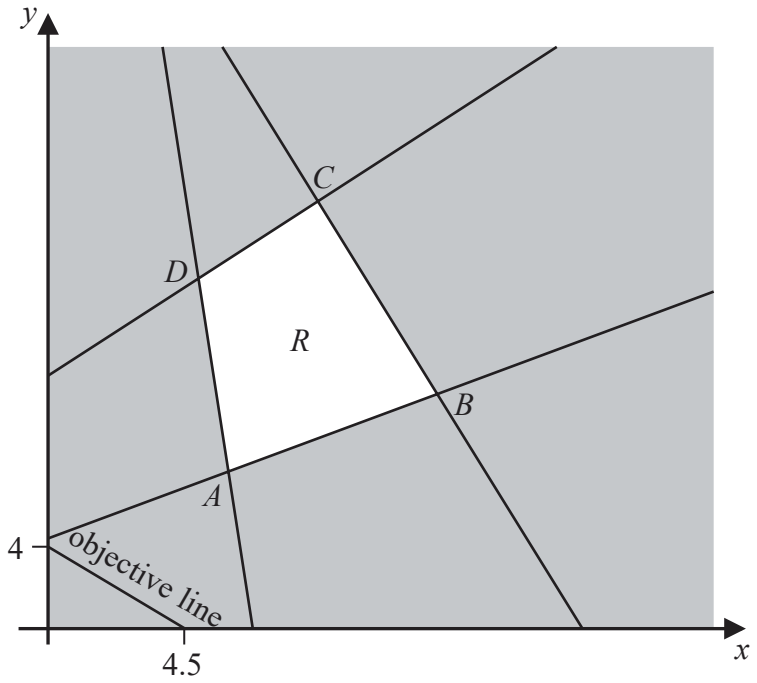


Figure 3

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Question 5 continued

(Total for Question 5 is 5 marks)

7. A farmer has 100 acres of land available that can be used for planting three crops: A, B and C.

It takes 2 hours to plant each acre of crop A, 1.5 hours to plant each acre of crop B and 45 minutes to plant each acre of crop C. The farmer has 138 hours available for planting.

At least one quarter of the total crops planted must be crop A.

For every three acres of crop B planted, at most five acres of crop C will be planted.

The farmer expects a profit of £160 for each acre of crop A planted, £75 for each acre of crop B planted and £125 for each acre of crop C planted.

The farmer wishes to maximise the profit from planting these three crops.

Let x, y and z represent the number of acres of land used for planting crop A, crop B, and crop C respectively.

(a) Formulate this information as a linear programming problem. State the objective, and list the constraints as simplified inequalities with integer coefficients.

(6)

The farmer decides that all 100 acres of available land will be used for planting the three crops.

(b) Explain why the maximum total profit is achieved when $-7x + 10y$ is minimised.

(2)

The farmer's decision to use all 100 acres reduces the constraints of the problem to the following:

$$x \geqslant 25$$

$$3x + 8y \geqslant 300$$

$$x + y \leqslant 100$$

$$5x + 3y \leqslant 252$$

$$y \geqslant 0$$

(c) Represent these constraints on Diagram 1 in the answer book. Hence determine, and label, the feasible region, R.

(4)

(d) (i) Determine the exact coordinates of each of the vertices of R.

(ii) Apply the vertex method to determine how the 100 acres should be used for planting the three crops.

(iii) Hence find the corresponding maximum expected profit.

(5)

(Total for Question 7 is 17 marks)

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Question 7 continued

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Question 7 continued

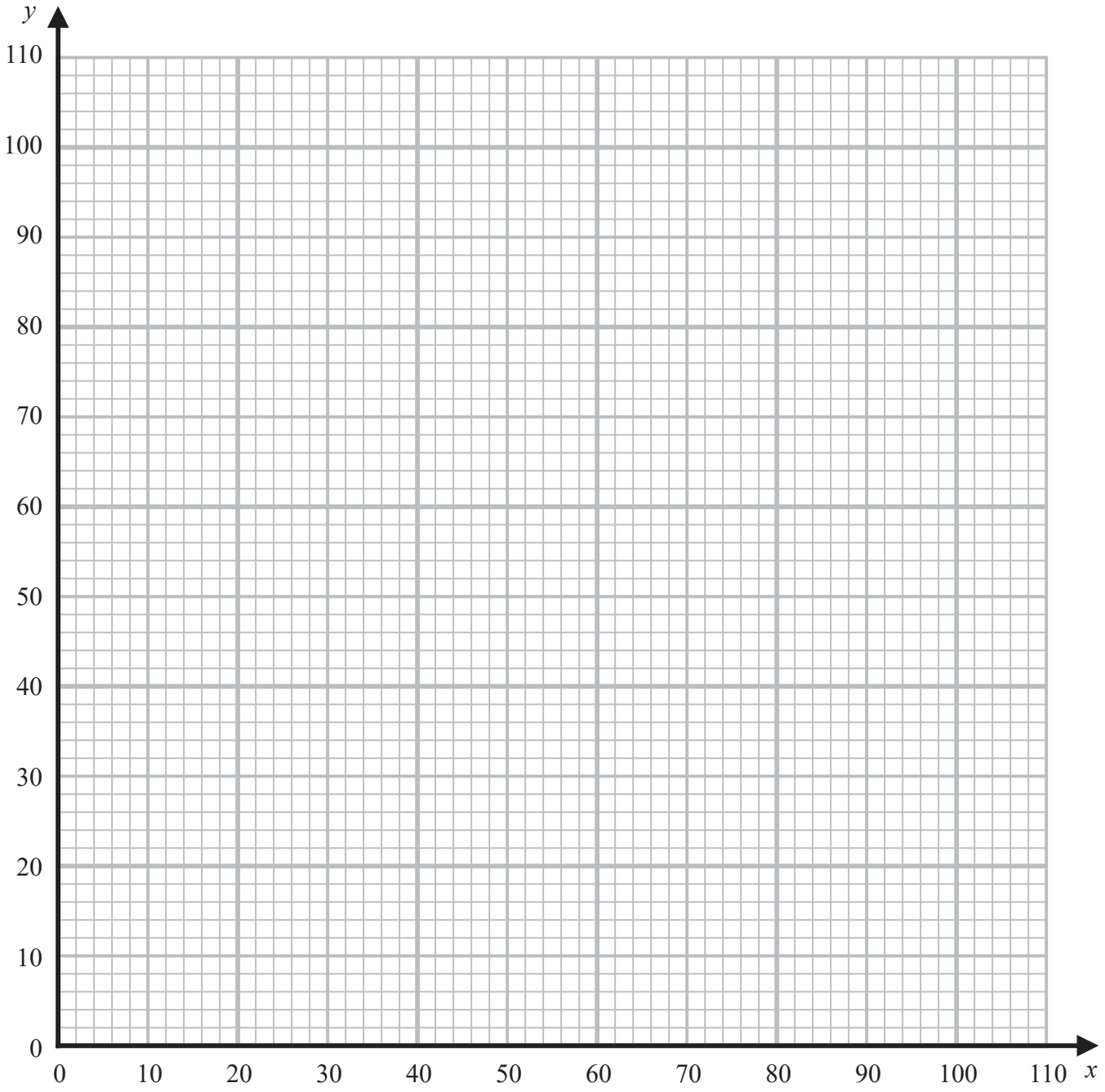


Diagram 1

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Question 7 continued

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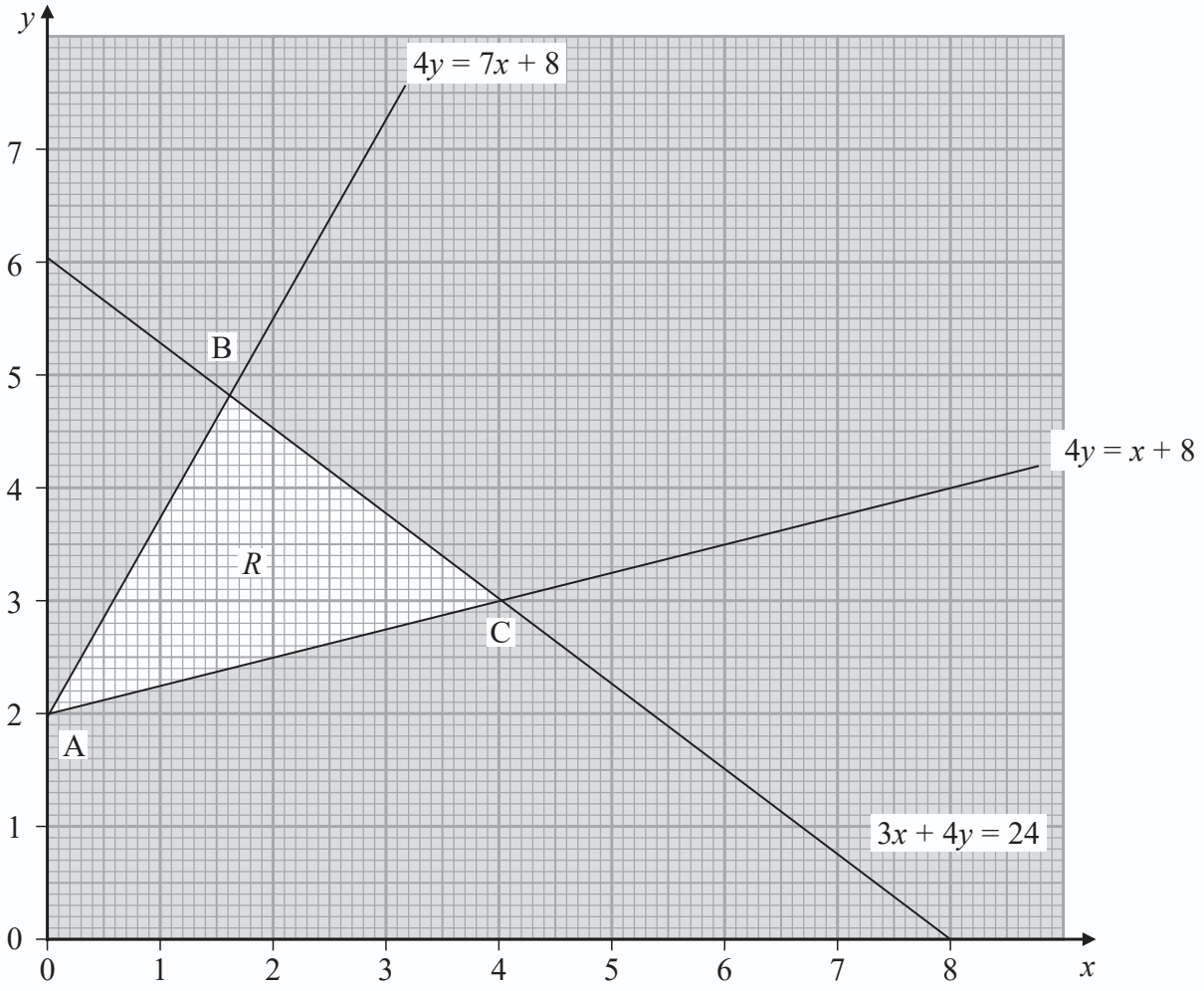


Figure 2

The graph in Figure 2 is being used to solve a linear programming problem in x and y. The three constraints have been drawn on the graph and the rejected regions have been shaded out. The three vertices of the feasible region R are labelled A, B and C.

(a) Determine the inequalities that define R.

(2)

The objective function, P, is given by

$$P = ax + by$$

where a and b are positive constants.

The minimum value of P is 8 and the maximum value of P occurs at C.

(b) Find the range of possible values of a. You must make your method clear.

(5)

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Question 6 continued

Q6

(Total 7 marks)

8. A bakery makes three types of doughnut. These are ring, jam and custard. The bakery has the following constraints on the number of doughnuts it must make each day.

- The total number of doughnuts made must be at least 200
- They must make at least three times as many ring doughnuts as jam doughnuts
- At most 70% of the doughnuts the bakery makes must be ring doughnuts
- At least a fifth of the doughnuts the bakery makes must be jam doughnuts

It costs 8 pence to make each ring doughnut, 10 pence to make each jam doughnut and 14 pence to make each custard doughnut. The bakery wants to minimise the total daily costs of making the required doughnuts.

Let x represent the number of ring doughnuts, let y represent the number of jam doughnuts and let z represent the number of custard doughnuts the bakery makes each day.

(a) Formulate this as a linear programming problem stating the objective and listing the constraints as simplified inequalities with integer coefficients.

(6)

On a given day, instead of making **at least** 200 doughnuts, the bakery requires that **exactly** 200 doughnuts are made. Furthermore, the bakery decides to make the minimum number of jam doughnuts which satisfy all the remaining constraints.

Given that the bakery still wants to minimise the total cost of making the required doughnuts, use algebra to

(b) (i) calculate the number of each type of doughnut the bakery will make on that day,

(ii) calculate the corresponding total cost of making all the doughnuts.

(5)

(Total 11 marks)

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Question 8 continued

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Question 8 continued

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4. A linear programming problem in x, y and z is described as follows.

Maximise $P = -x + y$

subject to

$$x + 2y + z \leqslant 15$$

$$3x - 4y + 2z \geqslant 1$$

$$2x + y + z = 14$$

$$x \geqslant 0,\ y \geqslant 0,\ z \geqslant 0$$

(a) (i) Eliminate z from the first two inequality constraints, simplifying your answers.

(ii) Hence state the maximum possible value of P

(4)

Given that P takes the maximum possible value found in (a)(ii),

(b) (i) determine the maximum possible value of x

(ii) Hence find a solution to the linear programming problem.

(3)

(Total for Question 4 is 7 marks)

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Question 4 continued

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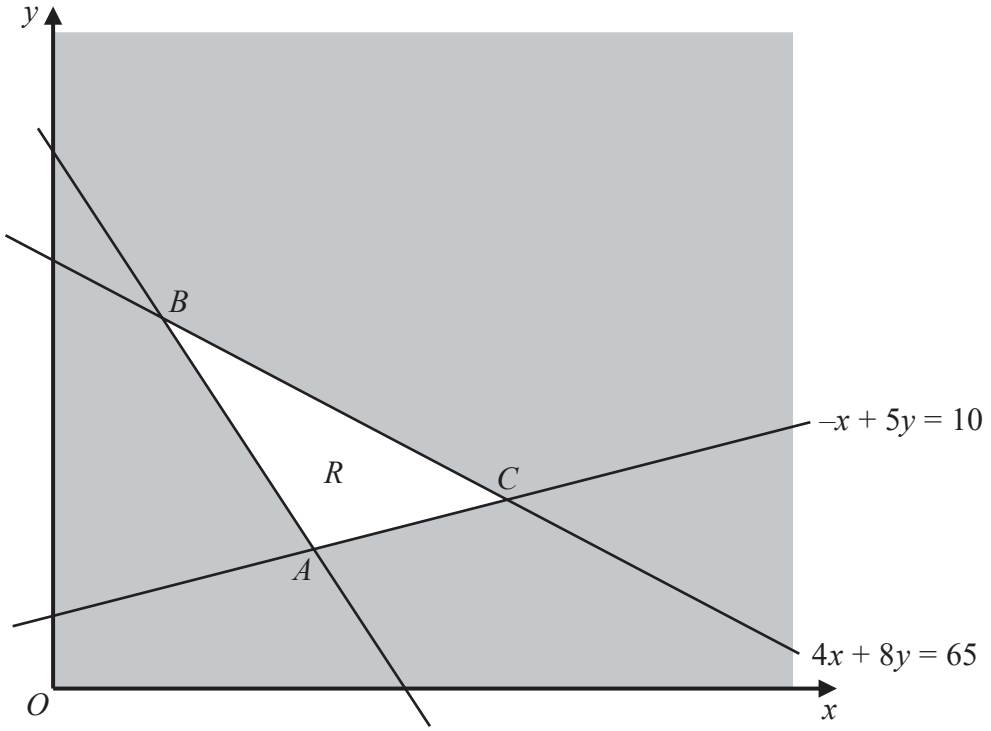


Figure 5

Figure 5 shows the constraints of a linear programming problem in x and y, where R is the feasible region. The equations of two of the lines and the three intersection points, A, B and C, are shown. The coordinates of C are $\left(\frac{35}{4}, \frac{15}{4}\right)$

The objective function is $P = x + 3y$

When the objective is to maximise $x + 3y$, the value of P is 24

When the objective is to minimise $x + 3y$, the value of P is 10

(a) (i) Find the coordinates of A and B.

(ii) Determine the inequalities that define R.

(8)

An additional constraint, $y \geqslant kx$, where k is a positive constant, is added to the linear programming problem.

(b) Determine the greatest value of k for which this additional constraint does not affect the feasible region.

(2)

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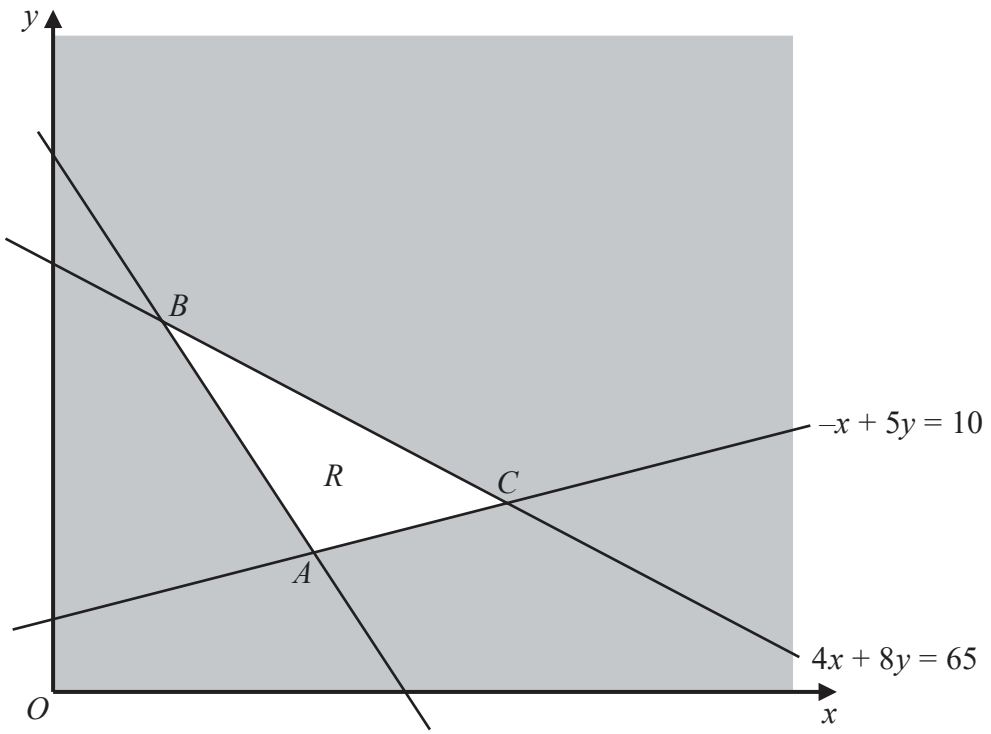


Figure 5

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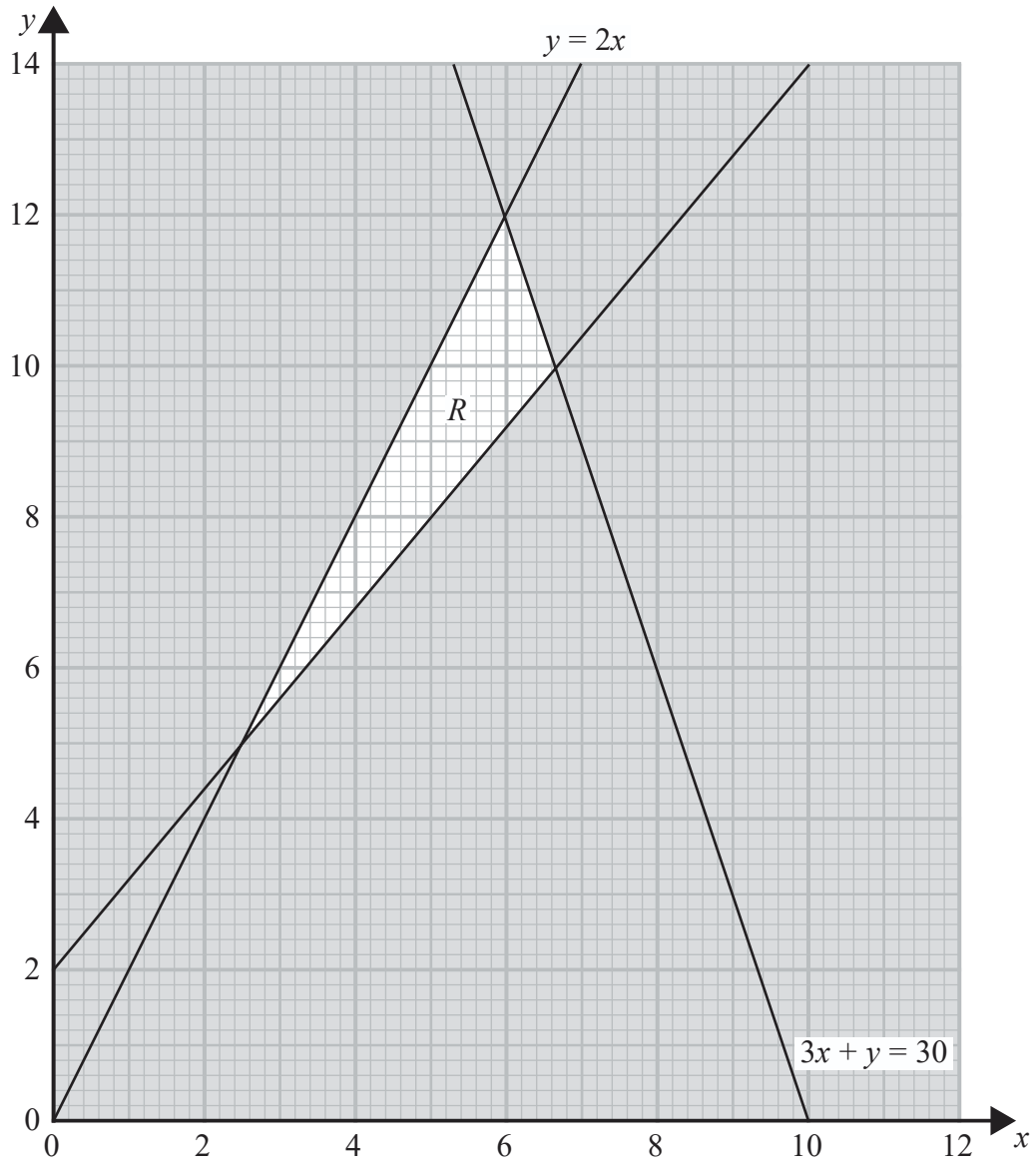


Figure 2

Figure 2 shows the constraints of a linear programming problem in x and y, where R is the feasible region. The equations of two of the lines are shown on the graph.

(a) Determine the inequalities that define the feasible region.

(3)

(b) Find the exact coordinates of the vertices of the feasible region.

(3)

The objective is to maximise P, where $P = 2x + ky$

(c) For the case $k = 3$, use the point testing method to find the optimal vertex of the feasible region and state the corresponding value of P.

(3)

(d) Determine the range of values for k for which the optimal vertex found in (c) is still optimal.

(2)

(Total for Question 4 is 11 marks)

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Question 4 continued

(Total for Question 4 is 11 marks)

8. A headteacher is deciding how to allocate prizes to the students who are leaving at the end of the school year.

There are three categories of prize: academic, sport, and leadership.

- Each academic prize costs £14, each sport prize costs £8, and each leadership prize costs £12. The total amount available to spend on all prizes is £976
- For every 5 academic prizes there must be at least 2 leadership prizes
- At least half the prizes must be academic
- 20% of the prizes must be for sport

The headteacher wishes to maximise the total number of prizes.

Let x, y and z represent the number of academic, sport and leadership prizes respectively.

(a) Formulate this as a linear programming problem in x and y only, stating the objective and listing the constraints as simplified inequalities with integer coefficients.

(8)

Given that the headteacher awards 16 sport prizes,

(b) calculate the corresponding number of leadership prizes that the headteacher awards. You must show your working.

(2)

(Total for Question 8 is 10 marks)

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Question 8 continued

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5. The head of a Mathematics department needs to order three types of paper. The three types of paper are plain, lined and graph.

All three types of paper are sold in reams. (A ream is 500 sheets of paper.)

Based on the last academic year the head of department formed the following constraints.

- At least half the paper must be lined
- No more than 15% of the paper must be graph paper
- The ratio of plain paper to graph paper must be 5 : 2

The cost of each ream of plain, lined and graph paper is £5, £12 and £15 respectively. The head of department has at most £834 to spend on paper.

The head of department wants to maximise the total number of reams of paper ordered.

Let x, y and z represent the number of reams of plain paper, lined paper and graph paper ordered respectively.

(a) Formulate this information as a linear programming problem in x and y only, stating the objective and listing the constraints as simplified inequalities with integer coefficients.

(7)

The head of department decides to order exactly 42 reams of lined paper and still wishes to maximise the total number of reams of paper ordered.

(b) Determine

(i) the total number of reams of paper to be ordered,

(ii) the number of reams of graph paper to be ordered.

(3)

(Total for Question 5 is 10 marks)

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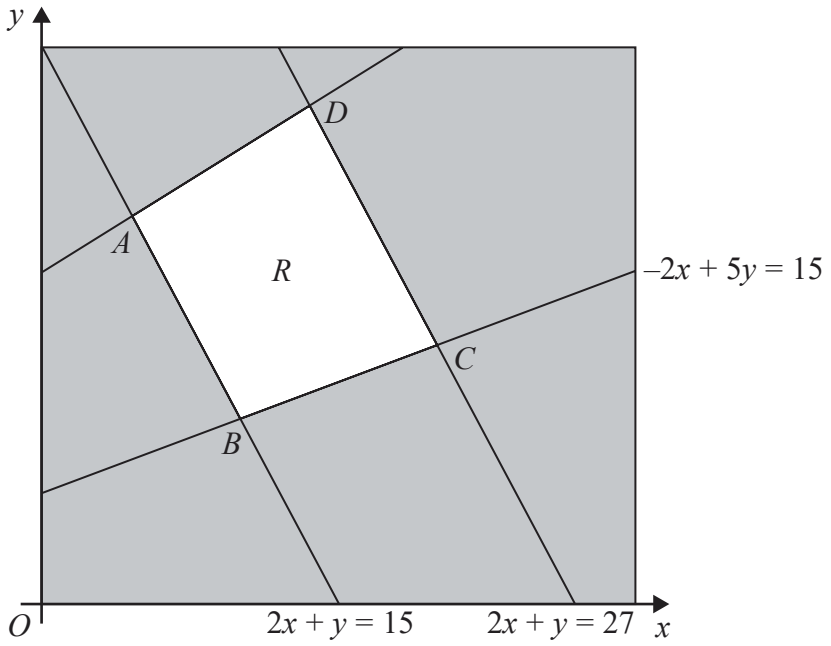


Figure 4

Figure 4 shows the constraints of a linear programming problem in x and y, where R is the feasible region.

The equations of three of the lines, and the four intersection points, A, B, C and D, are shown.

The four vertices of R are $A\left(\frac{9}{4},\frac{21}{2}\right)$, $B(5,5)$, $C(10,7)$ and $D\left(\frac{27}{4},\frac{27}{2}\right)$.

(a) Determine the inequalities that define R.

(4)

The objective function, P, is given by

$$P = 2x + ky$$

where k is a positive constant.

Given that the minimum value of P is at least half of the maximum value of P,

(b) determine the range of possible values of k.

(5)

(Total for Question 7 is 9 marks)

TOTAL FOR PAPER IS 75 MARKS

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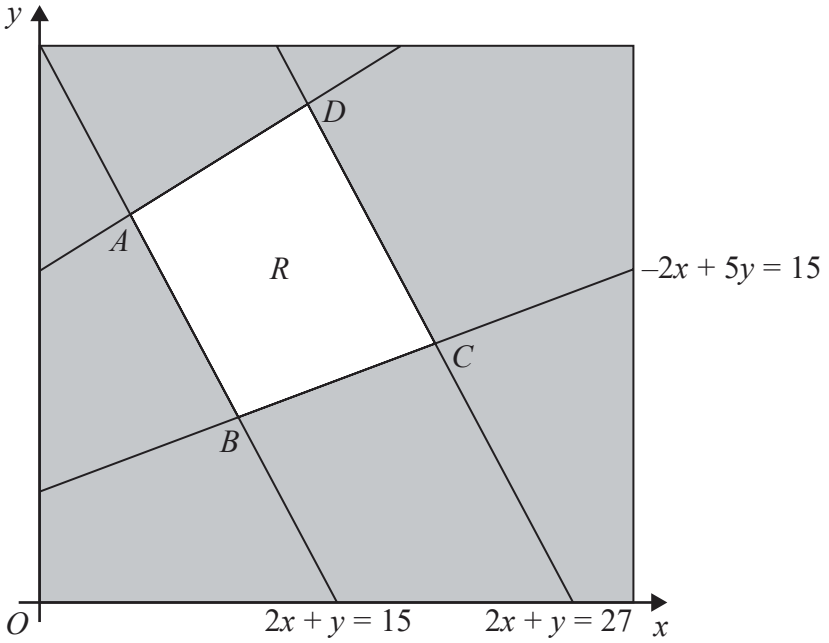


Figure 4

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Question 7 continued

Question 7 continued

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2. Chris has been asked to design a badge in the shape of a triangle XYZ subject to the following constraints.

- Angle Y should be at least three times the size of angle X
- Angle Z should be at least 50° larger than angle X
- Angle Y must be at most 120°

Chris has been asked to maximise the sum of the angles X and Y.

Let x be the size of angle X in degrees.

Let y be the size of angle Y in degrees.

Let z be the size of angle Z in degrees.

Formulate this information as a linear programming problem in x and y **only**. State the objective and list the constraints as simplified inequalities with integer coefficients.

You are **not** required to solve this problem.

(6)

(Total 6 marks)

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Question 2 continued

(Total 6 marks)

Q2

6. A linear programming problem in x and y is described as follows.

Maximise $P = kx + y$, where k is a constant

subject to: $3y \geqslant x$

$x + 2y \leqslant 130$

$4x + y \geqslant 100$

$4x + 3y \leqslant 300$

(a) Add lines and shading to Diagram 1 in the answer book to represent these constraints. Hence determine the feasible region and label it R.

(4)

(b) For the case when $k = 0.8$

(i) use the objective line method to find the optimal vertex, V, of the feasible region. You must draw and label your objective line and label vertex V clearly.

(2)

(ii) calculate the coordinates of V and hence calculate the corresponding value of P at V.

(3)

Given that for a different value of k, V is not the optimal vertex of R,

(c) determine the range of possible values for k. You must make your method and working clear.

(4)

(Total 13 marks)

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6.

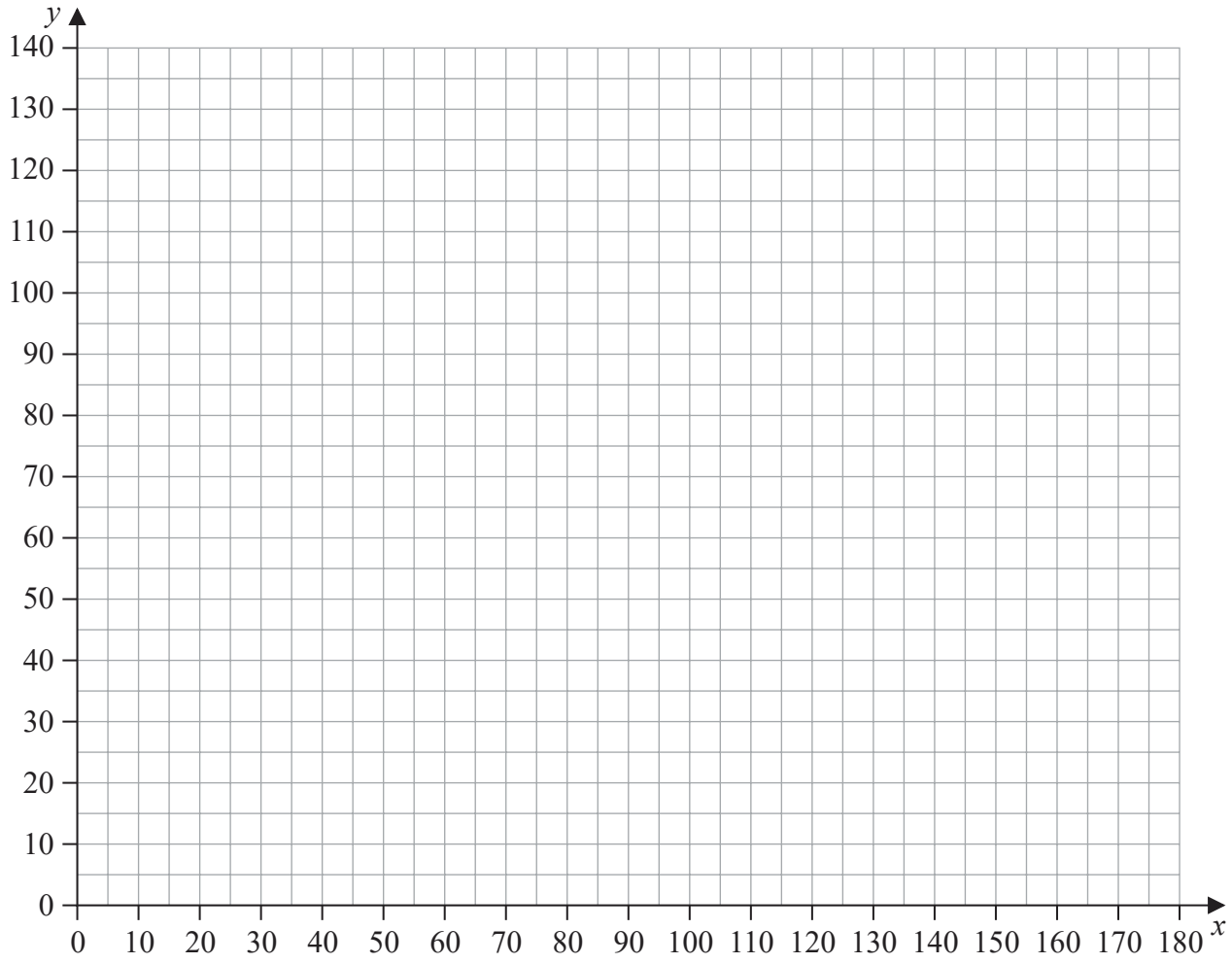


Diagram 1

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Question 6 continued

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Question 6 continued

(Total 13 marks)

Q6